

Youngjae Choi

Contact Information

Nationality: Dual Citizenship: Republic of Korea and United States
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Education

M.S. program in Department of Digital Media *Mar. 2024 - Present*
Soongsil University, Seoul, Korea
Advisor: Prof. Heewon Kim

B.S. in Department of Statistics and Actuarial Science *Mar. 2015 - Feb. 2022*
Soongsil University, Seoul, Korea

Projects

- **AI-powered XR Smart Space Solution Development** *SME R&D Support Program*
Participated in a national R&D project aimed at reconstructing interactive virtual spaces in AR by leveraging 3D Gaussian Splatting (3DGS). Developed a system that automatically estimates box colliders from reconstructed geometry, enabling real-time interaction within Unity-based XR environments.

Experience

Research Intern *Jun. 2023 - Feb. 2024*
Reality Lab, Soongsil University, Seoul, Korea *Mar. 2024 - Feb. 2026*

Teaching experience

Teaching Assistant for *Deep learning* at Soongsil University (spring 2024)
Teaching Assistant for *Machine learning* at Soongsil University (Spring 2025)

Working experience

Motifdrive(Autonomous Vehicle Start-up) *Jan. 2026 -*
Researched generative AI for synthetic data generation and augmentation in end-to-end autonomous driving, developed VLM-based data management and scenario mining frameworks, and contributed to 3D object detection research using multi-modal sensor data.

Research interests

Computer vision, Deep learning, Motion Generation, Image Generation, Embodied AI

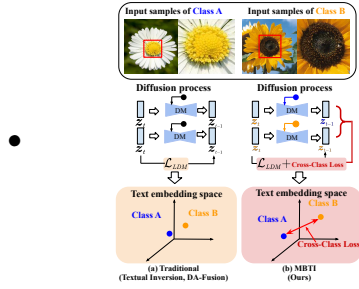
Research Highlight

Conducting research on multimodal generative models with a focus on generating fine-grained images and expressive human motions from text inputs. This includes developing methods that reflect detailed semantics and emotional context in the generation process.

Skills

Python, PyTorch, MATLAB, Object Detection & Segmentation(MMCV), 3DGS, L^AT_EX

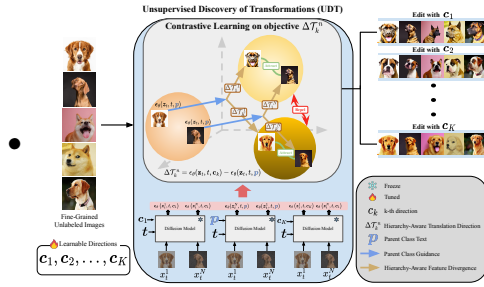
Publications



MBTI: Metric-Based Textual Inversion for Fine-Grained Image Generation

ByungKwan Chae*, **Youngjae Choi***, and Heewon Kim

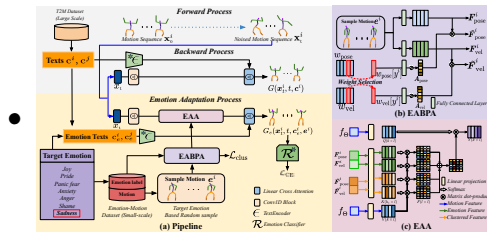
WACV 2026



UDT: Unsupervised Discovery of Transformations between Fine-Grained Classes in Diffusion Models

Youngjae Choi*, Hyunsuh Koh*, Byungkwan Chae*, Hojae Jeong*, and Heewon Kim

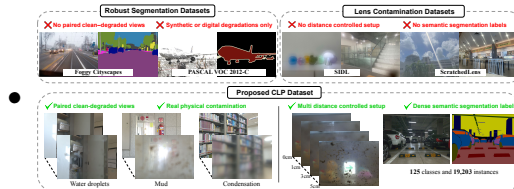
BMVC 2025



FEEL: Focus on Emotion-Enriched Motion Using Unsupervised Learning for Text-to-Motion Generation

Youngjae Choi*, Hyunsuh Koh, and Heewon Kim

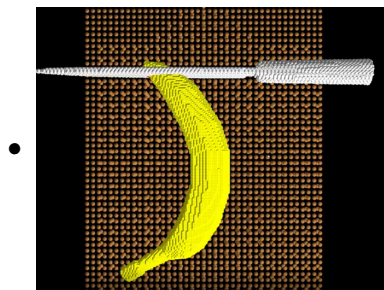
CVPRW 2026



CLP: A Real-World Dataset of Contaminated Lens Protectors for Robust Semantic Segmentation

Sungyong Park*, Sooyoung Choi*, Hyunseo Koh, **Youngjae Choi**, and Heewon Kim

CVPR 2026



CulinaryCut: A Physics-Grounded Cutting Benchmark for Vision-Language-Action Models

Hyunseo Koh*, **Youngjae Choi***, Chang-Yong Song*, Misa Viveiros, David Hyde, Heewon Kim

ECCV 2026

References

Advisor Heewon Kim
Professor
Soongsil University
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<https://sites.google.com/view/kimheewon/>